Circular Motion

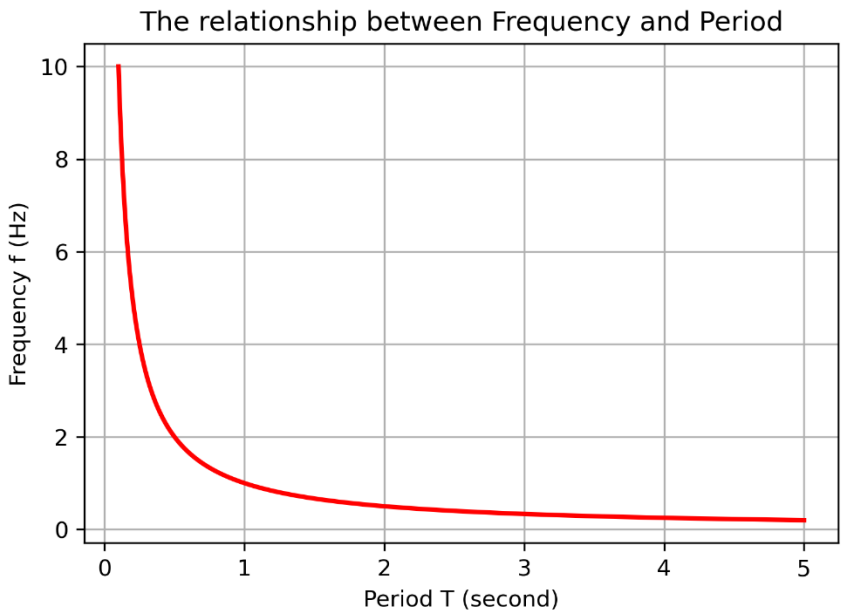
Circular motion holds a special place among the types of motion in physics. It is one of the fundamental topics of classical mechanics and helps us understand many phenomena in nature. Circular motion is defined as the motion of an object along a circle with a constant radius. For example, the Earth rotating about its own axis, planets orbiting the Sun, or the rotation of a wind turbine's blades are all examples of circular motion.

Circular motion is divided into two main categories: uniform circular motion and non-uniform circular motion. If an object covers equal distances in equal time intervals along a circular path, its motion is called *uniform circular motion*.

To analyze uniform circular motion, several fundamental concepts must be understood. For instance, the **period (T)** is the time it takes an object to complete one full revolution. The **frequency (f)** is the number of revolutions per second. There is an inverse relationship between these two quantities, expressed by:

$T=1/f$

Figure 1: Frequency–Period Graph



The **linear (tangential) speed** of an object performing uniform circular motion is the distance covered per unit time and is always tangent to the path. Since the object travels a distance equal to the circumference of the circle in one period, the linear speed vvv is given by:

$v=2\pi.r.T$

Another important concept is **angular velocity**. Angular velocity is the angle swept by the radius vector per unit time. When an object completes one full revolution, it sweeps an angle of 2π radians. Angular velocity, denoted by ω omega is expressed as:

$\omega=2\pi/T$

There is also a relationship between angular velocity and linear speed:

$v=\omega.r$

In uniform circular motion, although the magnitude of the velocity remains constant, its **direction** continuously changes. Therefore, the acceleration is not zero; it is always directed toward the center of the circle. This acceleration is called **centripetal acceleration**, denoted by a:

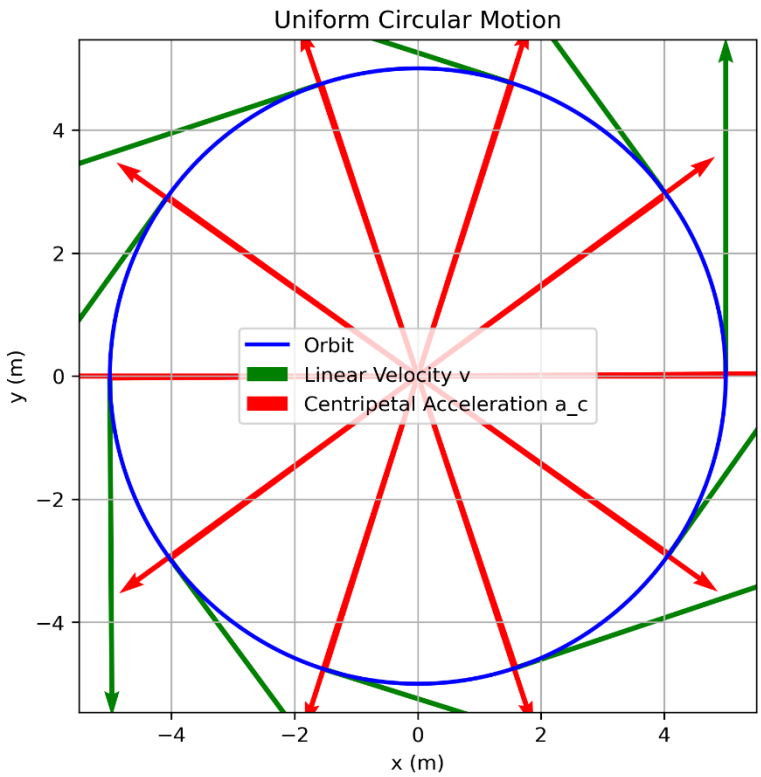
$a=v^{2/}r$

The force that causes this directional change—always pointing toward the center and perpendicular to the velocity vector—is called the **centripetal force**, expressed as:

$F=m.v^2/r$

For example, in order for a car to safely take a turn, the friction between the tires and the road must provide the required centripetal force. If the frictional force is insufficient, the vehicle will skid outward.

Figure 2: Uniform Circular Motion Graph



In **non-uniform circular motion**, the magnitude of the velocity changes, unlike in uniform circular motion. In this type of motion, the object does not rotate with a constant angular

velocity; instead, its speed increases or decreases. Due to the net forces acting on the object, the acceleration has two components: **centripetal acceleration** and **tangential acceleration**.

Centripetal acceleration keeps the object on its circular path and always points toward the center. **Tangential acceleration**, on the other hand, changes the magnitude of the object's speed. If the object speeds up, the tangential acceleration is in the same direction as the velocity; if it slows down, it is in the opposite direction.

The total acceleration in non-uniform circular motion is the vector sum of these two components. For this reason, both the direction and the magnitude of the velocity change simultaneously.

Many real-life examples involve non-uniform circular motion. For instance, a car turning while accelerating or braking is an example of such motion because the car both changes direction and alters its speed.

In conclusion, circular motion is one of the most important types of motion in nature. With its uniform and non-uniform forms, it has wide applications in vehicle dynamics, engineering, and physical simulations.

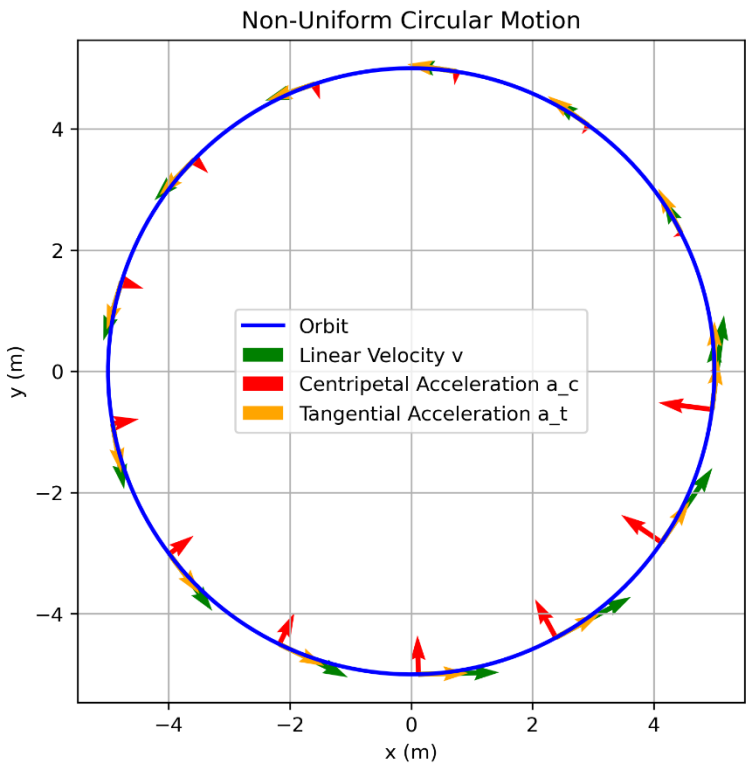



Figure 3: Non-Uniform Circular Motion Graph